

Energy-dispersive X-ray diffraction measurements

1. Overview

Energy dispersive X-ray diffraction measurements were conducted on an AM Bench 2018 IN625 test artifact produced using laser powder-bed fusion additive manufacturing. The measurements were made on the ID1A3 beamline at the Cornell High Energy Synchrotron Source (CHESS). The incident X-ray beam was masked to a 0.25 mm × 0.25 mm cross-section by a pair of incident beam slits. The beam was diffracted through two additional sets of 0.25 mm slits with a fixed 2θ angle of 9.18°. The multiple slit geometry results in a 'rhomboidal' shaped diffraction volume of 0.25 mm × 1.5 mm × 0.25 mm, where the long dimension is along the Y dimension of the sample. Diffraction data were collected over the X-Z plane at the mid-point in Y. The measurement locations for the lattice strains are illustrated in Figure 1.

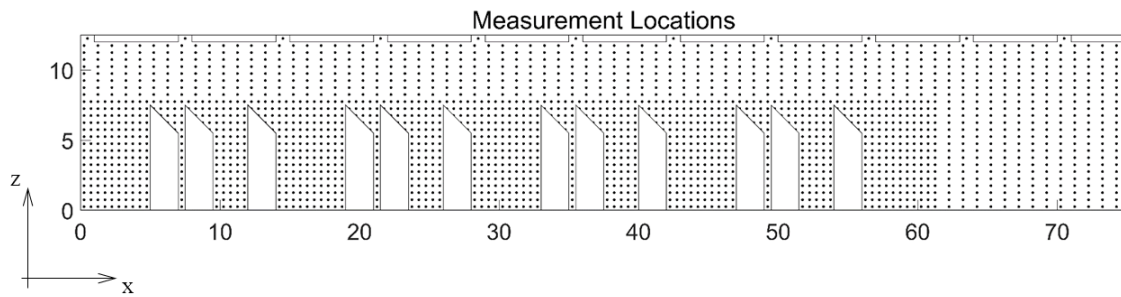


Figure 1: Measurement locations for the residual strains measured on the ID1A3 beamline at CHESS.

2. Summary Results

Figure 2 shows the lattice strain components of the strain along the (a) build direction (Z), (b) the longitudinal strain (X), (c) the Z-X strain at 45° and (d) the shear strain (γ_{xz}). The γ_{xz} was calculated by using the equation $\gamma_{xz} = \frac{1}{2} (\epsilon_{xx} + \epsilon_{zz}) - \epsilon_{zx}$.

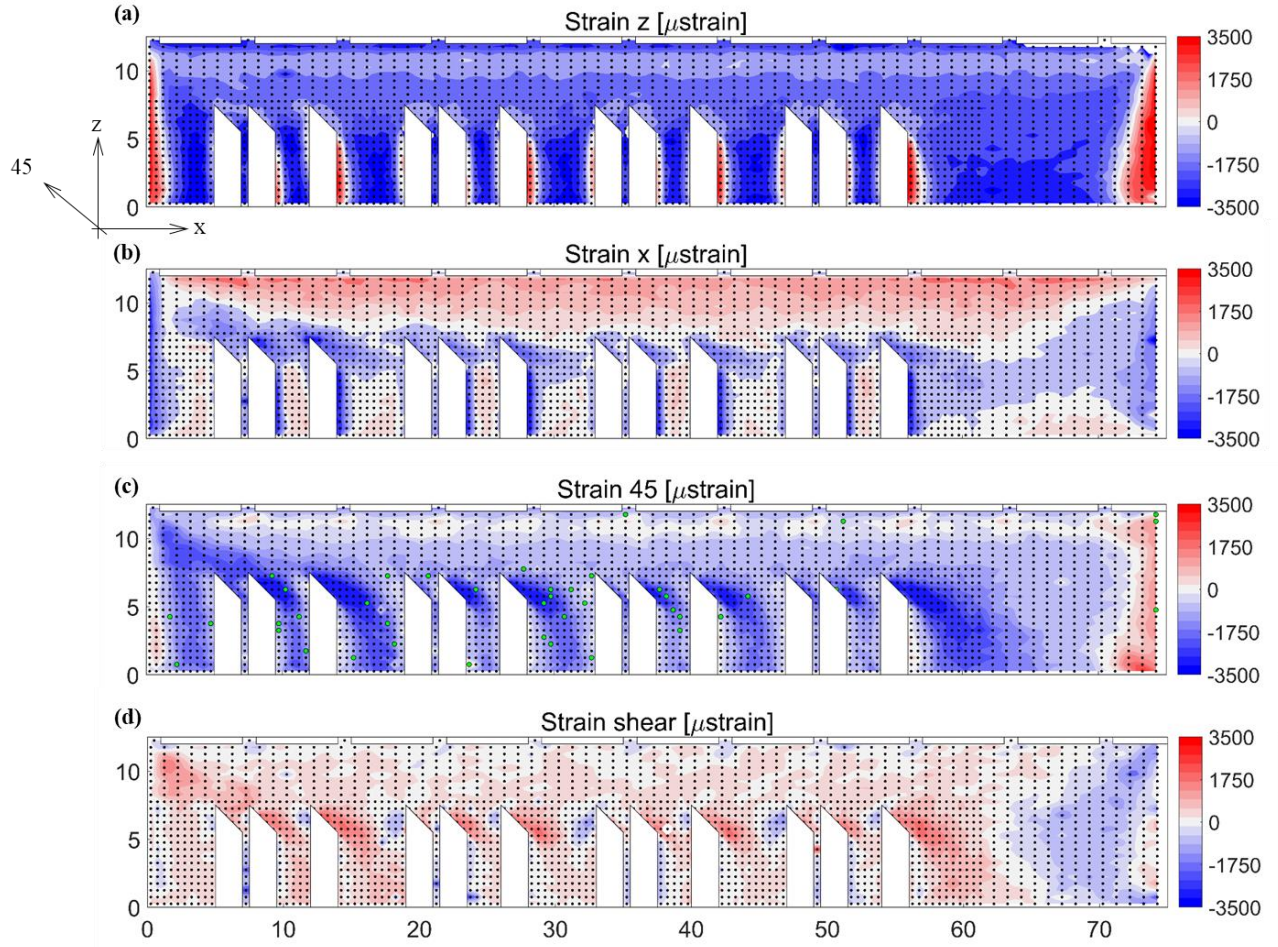


Figure 2: Residual strain maps over the investigated middle cross-section of the sample ($Y = 2.5$ mm) of the (a) strain along the build direction (Z), (b) the longitudinal strain (X), (c) the Z_X strain at 45° and (d) the shear strain component (γ_{xz}). The X and Z positions are shown in mm.

3. Data Files

The lattice strain components along the X , Z and the $Z-X$ at 45° for the IN625 specimen are presented in a data table as a tab-delimited text file. The text file contains six columns, indicating the X position, the Z position, the ϵ_{xx} , the ϵ_{zz} , the ϵ_{zx} (at 45°) and the γ_{xz} . The X and Z positions indicate the location (in mm) of the **center** of the measurement gauge volume, with the origin set at the bottom left corner of the $X-Z$ plane. Additionally, each of the results plots in Figure 2 are provided as high resolution Portable Network Graphics (*.png) image files, and Matlab figure (*.fig) files.